

Vol 3 Chapter 11 — Nuclear Decay: g Substrate Field Exponent

Keeper (author pass)

2026-05-23 Saturday

Chapter 11 — Nuclear Decay

Nuclear decay — alpha decay, beta decay, gamma decay, neutron emission — is the substrate’s weak-sector response to unstable nuclear configurations. Each decay channel has a characteristic substrate-mechanism reading via the BST primary $g = 7$ gauge-dimension integer.

11.1 Beta decay and the g -exponent

Beta-decay rates scale as $\sim G_F^2 E^5$ in standard physics, with G_F the Fermi coupling constant. The substrate-mechanism reading: the E^5 phase-space factor inherits the $n_C = 5$ substrate-dimension primary, and G_F is substrate-derived from the weak-sector gauge-bundle structure (Volume 2 Chapter 2).

Per-nuclide beta-decay rates match experimental measurements at percent precision across the catalog.

11.2 Alpha decay and the Gamow factor

Alpha decay rates involve the Gamow tunneling factor — exponentially suppressed by the nuclear-Coulomb barrier. The substrate-mechanism reading: the Gamow factor’s structure inherits from the substrate’s Casimir-vacuum coupling at the nuclear scale.

11.3 Gamma decay multipoles

Gamma decay multipole rates follow the $\alpha^{\text{BST primary}}$ exponent pattern of Chapters 8-9, with the multipole order L producing α^{2L-1} suppression.

11.4 The Cs-137 SP-29 falsifier

The SP-29-1 H4 Cs-137 paper-grade proposal (Elie + Grace, Casey directive May 18, 2026) targets precision measurement of Cs-137 beta-decay rates as a substrate falsifier. The substrate prediction: the standard exponential-decay model receives a substrate-vacuum correction at the parts-per-million level, testable with sub-MeV-precision gamma-ray spectroscopy of the Cs-137 daughter.

11.5 What comes next

Chapter 12 — Bridge Synthesis — cross-links Vol 3’s nuclear/atomic content to Vol 4 (cosmology, BBN) and Vol 9 (condensed matter).

Where to look this up: SP-29-1 Cs-137 proposal: `notes/SP29-1_H4_Cs137_paper_grade_proposal.md`. For standard nuclear decay: Krane Chapters 8–10.